MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEFSC200 - INSTRUMENTATION - CHEMICAL

(2024 ADMISSION ONWARDS)

Scheme of Evaluation

Part A
Answer in one word
Answer all questions
Each question carries 1 marks

1. Discrete quantized energy states of molecules associated with electronic, vibrational, and rotational motions-
2. Dynode
3. UV–Visible Detector / Fluorescence Detector / Refractive Index Detector
4. Split / Splitless / On-column injection
5. Double-beam dispersive spectrophotometer / Prism monochromator/ Grating
6. Absorbance is directly proportional to path length for a given concentration
7. Separation by liquid chromatography followed by mass-to-charge analysis in mass spectrometer
8. Isocratic / Gradient elution
9. Detection of volatile poisons / Alcohol analysis / Explosives analysis
10. Pollutant / Contaminant / Trace metal levels

Part B
Short Answer Questions
Answer any 5 questions
Each question carries 3 marks

11. **Source:** Provides stable, intense radiation across desired wavelengths (e.g., deuterium lamp for UV, Nernst glower for IR).
Monochromator: Disperses polychromatic light into individual wavelengths using grating/prism.
Detector: Converts light signal into electrical output proportional to intensity (photodiode, PMT).
→ Together they enable accurate measurement of absorbance and transmittance.
(1 mark each = 3)

12. Tandem MS involves two mass analyzers separated by a collision cell. The first MS selects a specific parent ion. Collision-induced dissociation breaks it into fragment ions. The second MS analyzes these fragments to identify structure.

Applications include drug profiling, protein sequencing, and toxicological screening. **(Principle–2 ; Applications–1)**

13. Typical solvents depend on analyte polarity:- Ethyl acetate, Toluene, Methanol / Hexane / Chloroform
    Solvent choice determines Rf value and resolution. **(Any three = 3 marks)**
14. Gas chromatography first separates volatile compounds based on partition between stationary and mobile phases. The effluent enters the mass spectrometer where molecules are ionized and sorted by mass-to-charge ratio. The combination provides both qualitative and quantitative identification. (Separation–1 ; Ionization–1 ; Identification–1)
15. The detector converts transmitted IR radiation into an electrical signal.
    Common detectors include Thermocouple, Golay cell, and DTGS.
    They measure intensity differences between sample and reference beams, producing absorbance spectra for functional group identification.
    (Working–2 ; Example–1)
16. Out of context- any attempt may give full marks
17. Mass spectrometry measures mass-to-charge ratio of ions.
    Sample molecules are ionized (electron impact or ESI), accelerated by electric fields, and separated in mass analyzers (quadrupole, TOF).
    Detector records intensity to produce a mass spectrum used for identification and quantification.
    (Ionization–1 ; Separation–1 ; Detection–1)

Part C
Short Essay Questions
Answer any 5 questions
Each question carries 6 marks

18. Detectors transform the separated components into measurable signals, increasing sensitivity and precision. Types and contributions: UV–Vis Detector: Measures absorbance of eluted species. Fluorescence Detector: Enhances selectivity for fluorescent compounds. Refractive Index Detector: Universal but less sensitive. Electron Capture Detector (GC): Highly sensitive for halogenated compounds. Mass spectrometric Detector: Provides molecular identity and quantitation. Detectors allow trace-level analysis in forensic samples like drugs, inks, and toxicants. (Definition–1; Types–3; Forensic importance–2)
19. Any relevant comparison on Principle, source, detector, advantage and application (6 marks)

Aspect	IR	FTIR	Raman
Principle	Absorption of IR radiation causing vibration	Interferometric measurement of IR spectrum	Inelastic scattering of monochromatic light
Source	Glower	Broadband + Interferometer	Laser
Detector	Thermal	DTGS/MCT	CCD
Advantage	Simple	High resolution, speed	Water-insensitive

Application	Organic group identification	Complex matrix analysis	Pigments, inks

20. Line width depends on magnetic field homogeneity and relaxation time (T). Broader lines: short relaxation, less resolution. Narrow lines: better separation of chemical shifts and coupling. Inter-nuclear interactions and viscosity also affect width. Understanding line width aids in accurate chemical shift and multiplet analysis for structural elucidation. (Definition–2; Factors–2; Importance–2)
21. Electronic transitions occur when molecules absorb UV–Vis radiation (200–800 nm).
    Types: $\sigma \rightarrow \sigma^*$, $n \rightarrow \sigma^*$, $\pi \rightarrow \pi^*$, $n \rightarrow \pi^*$.
    Transition energy depends on bond type and substituents.
    Chromophores and auxochromes shift absorption bands (hypsochromic or bathochromic shifts).
    Used to identify conjugation and concentration via Beer–Lambert Law.
    (Transitions–3; Influence–2; Application–1)
22. Not relevant question - any attempt may give full marks
23. High α and Rs values indicate better chromatographic efficiency.
    These parameters guide optimization of column length, stationary phase, and mobile phase.
    (Formulas–3; Meaning–2; Interpretation–1)
24. Spin quantum number (I) determines number of orientations (2I + 1).
    This forms basis for chemical shift and spin-spin coupling observed in spectra.
    (Definition–2; Theory–2; Relevance–2)

Part D- Essay Type Questions- Answer any 1 questions
Each question carries 15 marks

25. Electromagnetic radiation interacts with matter through absorption, emission, reflection, or scattering.
    Different frequencies produce distinct phenomena.

Region	Type of Interaction	Effect
Gamma / X-ray	Ionization	Ejection of electrons
UV–Visible	Electronic excitation	$n \rightarrow \pi^*$, $\pi \rightarrow \pi^*$ transitions
IR	Vibrational	Bond stretch and bend
Microwave	Rotational	Molecular rotation
Radio	Nuclear spin transitions	Basis of NMR

Marking (15): Concept–3; Table–5; Beer–Lambert–2; Selection rules–2; Applications–2; Conclusion–1.

26. In mass spectrometry, *scanning modes* determine how the instrument collects ion signals to analyze a sample. The two commonly used modes are **Full Scan** and **SIM (Selected Ion Monitoring)**.

Feature	Full Scan	SIM
m/z Range	Broad (50–500)	Few specific m/z values

Purpose	Qualitative analysis	Quantitative analysis
Sensitivity	Lower	Higher
Use Case	Screening unknowns	Targeted compound detection

Definition of both modes – 4 marks, Principles and working – 4 marks, Advantages/Limitations – 4 marks, Forensic application and conclusion – 3 marks